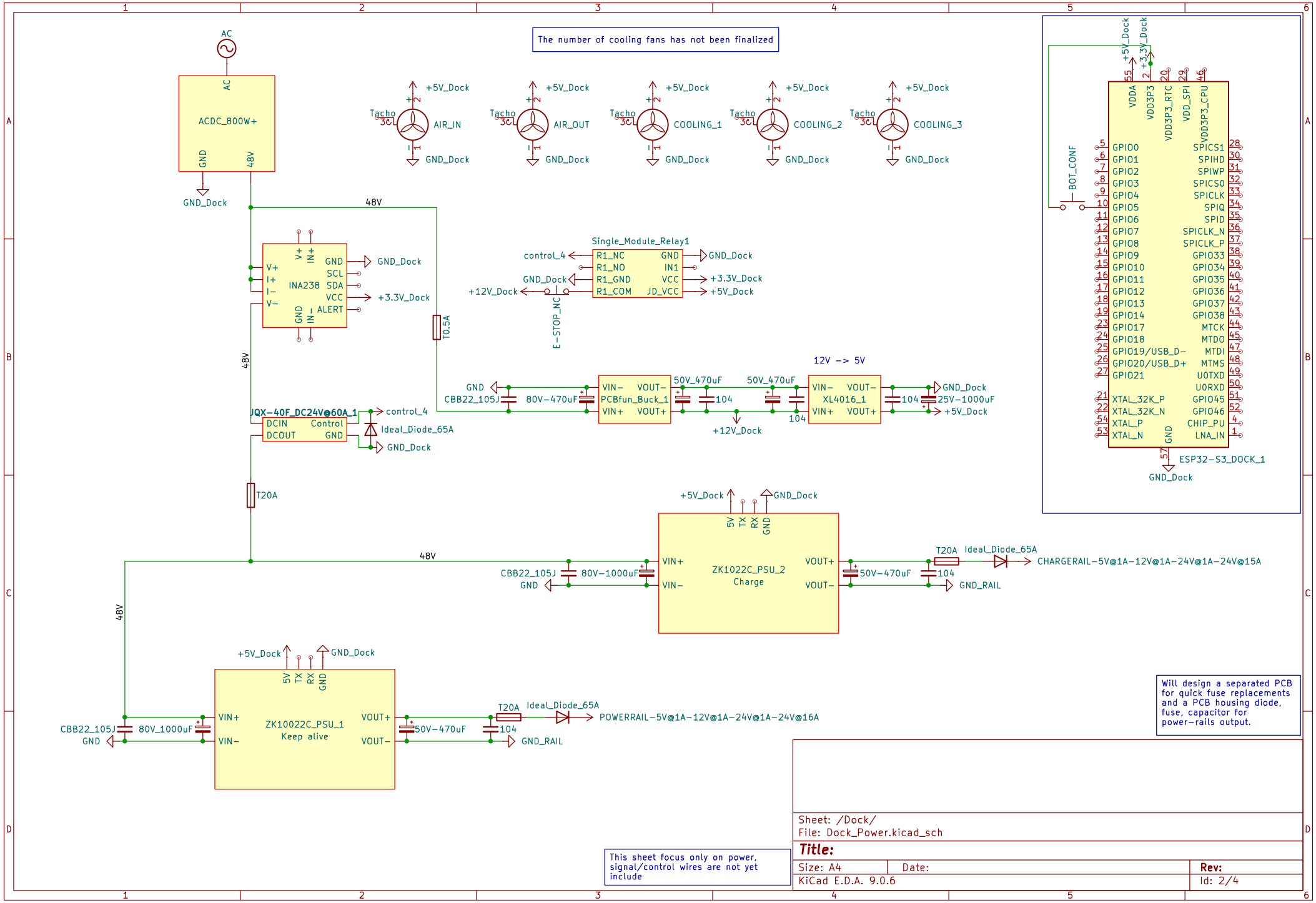
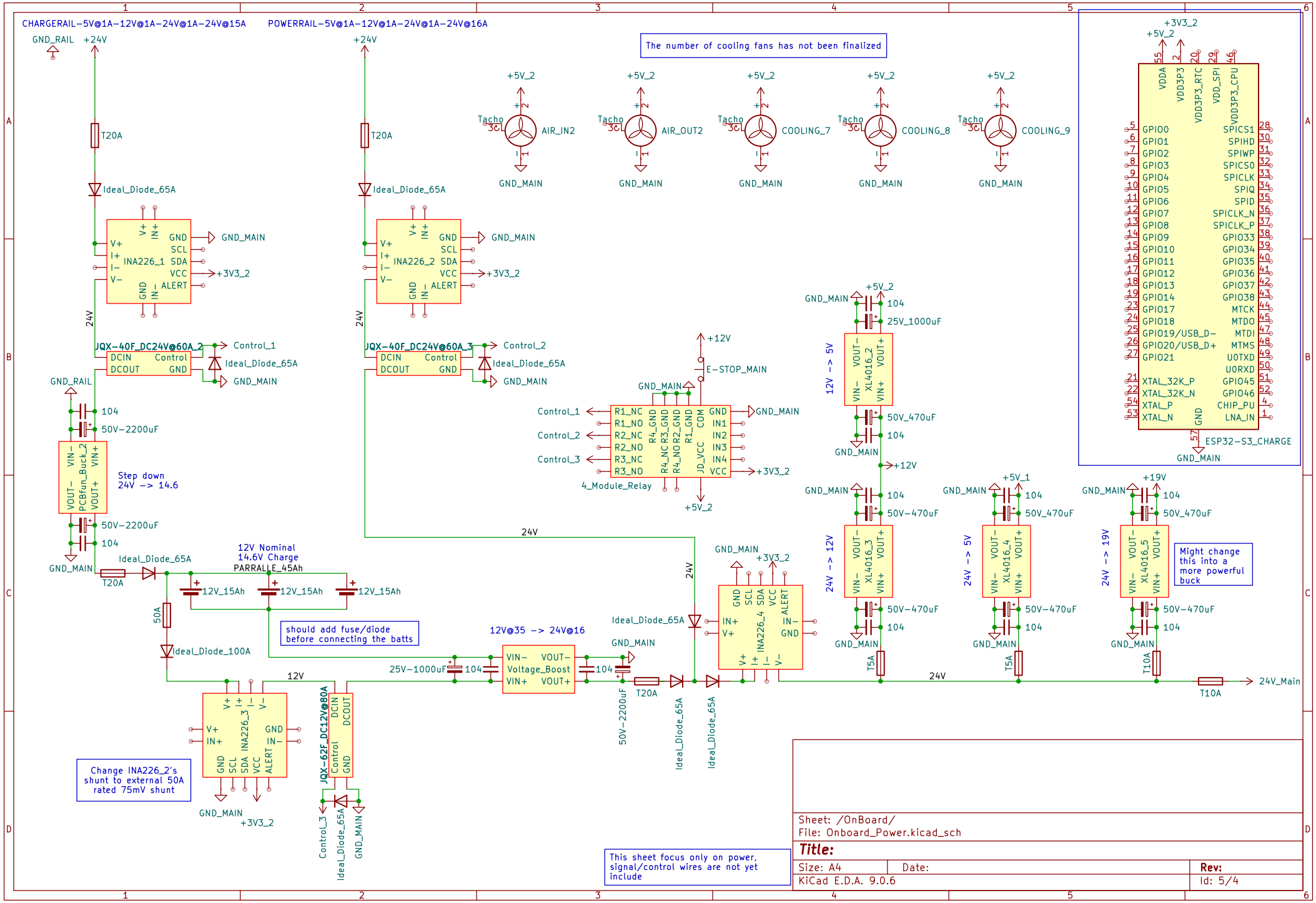
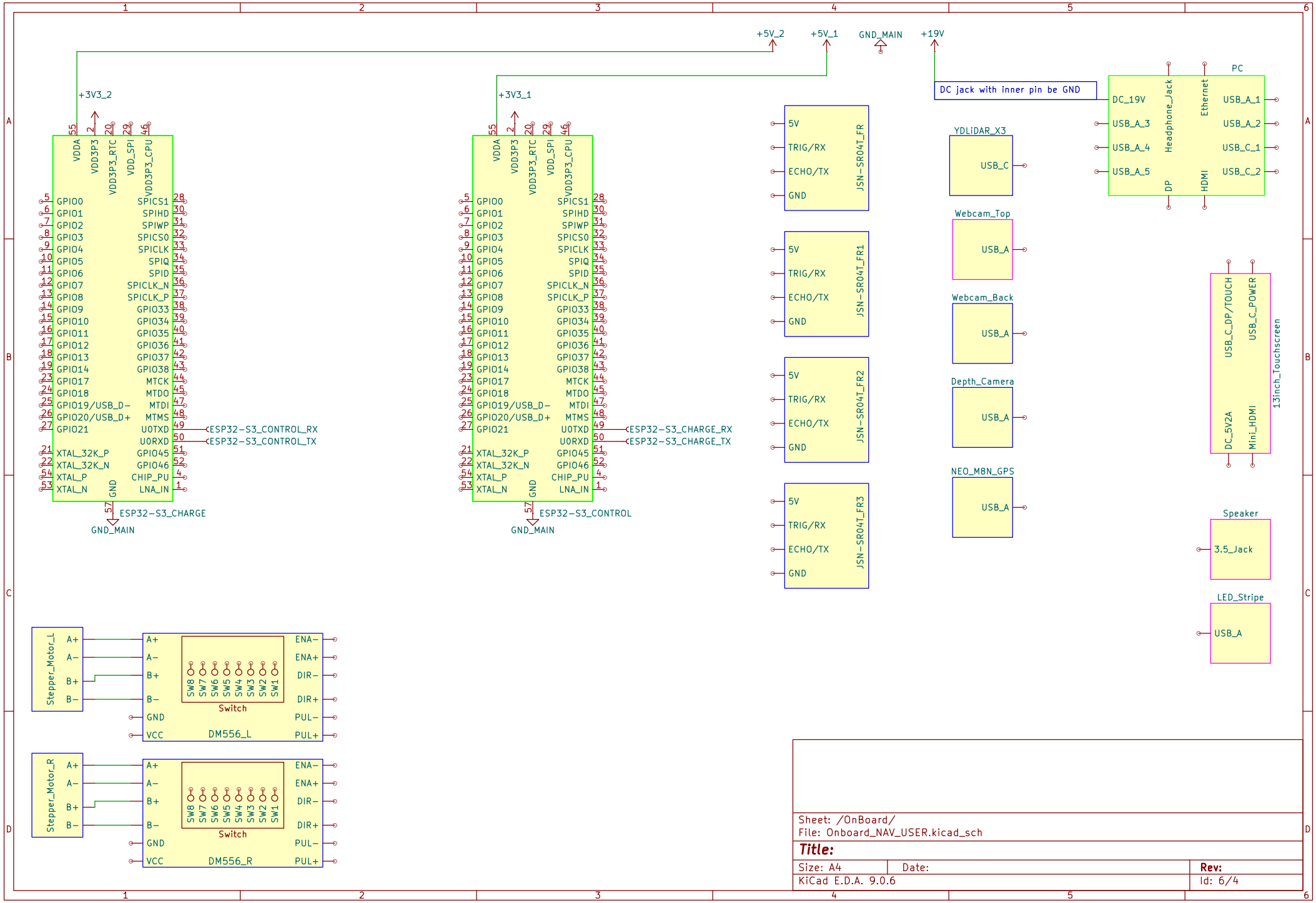

The number of cooling fans has not been finalized
AC
ACDC_800W+
GND
48V
GND_Dock
+5V_Dock
Tacho
AIR_IN
AIR_OUT
COOLING_1
COOLING_2
COOLING_3
INA238
V+
I+
I−
V−
SCL
SDA
VCC
ALERT
+3.3V_Dock
T0.5A
Single_Module_Relay1
control_4
R1_NC
R1_NO
R1_GND
R1_COM
GND
IN1
JD_VCC
+12V_Dock
E−STOP_NC
12V −> 5V
CBB22_105J
80V−470uF
PCBfun_Buck_1
50V_470uF
104
XL4016_1
25V−1000uF
JQX−40F_DC24V@60A_1
DCIN
DCOUT
Control
Ideal_Diode_65A
T20A
ESP32−S3_DOCK_1
BOT_CONF
ZK1022C_PSU_2
Charge
CHARGERAIL−5V@1A−12V@1A−24V@1A−24V@15A
GND_RAIL
ZK10022C_PSU_1
Keep alive
POWERRAIL−5V@1A−12V@1A−24V@1A−24V@16A
Will design a separated PCB for quick fuse replacements and a PCB housing diode, fuse, capacitor for power−rails output.
This sheet focus only on power, signal/control wires are not yet include
Sheet: /Dock/
File: Dock_Power.kicad_sch
Title:
Size: A4
Date:
Rev:
KiCad E.D.A. 9.0.6
Id: 2/4


CHARGERAIL-5V@1A-12V@1A-24V@1A-24V@15A
POWERRAIL-5V@1A-12V@1A-24V@1A-24V@16A
The number of cooling fans has not been finalized
GND_RAIL
+24V
T20A
Ideal_Diode_65A
INA226_1
INA226_2
GND_MAIN
+3V3_2
JQX-40F_DC24V@60A_2
JQX-40F_DC24V@60A_3
Control_1
Control_2
Control_3
E-STOP_MAIN
+12V
+5V_2
4_Module_Relay
AIR_IN2
AIR_OUT2
COOLING_7
COOLING_8
COOLING_9
Tacho
ESP32-S3_CHARGE
104
50V-2200uF
Step down
24V -> 14.6
12V Nominal
14.6V Charge
PARRALLE_45Ah
12V_15Ah
50A
Ideal_Diode_100A
should add fuse/diode
before connecting the batts
12V@35 -> 24V@16
Voltage_Boost
25V-1000uF
12V
24V
Change INA226_2's
shunt to external 50A
rated 75mV shunt
JQX-62F_DC12V@80A
Control_3
Ideal_Diode_65A
12V -> 5V
24V -> 12V
24V -> 5V
24V -> 19V
XL4016_2
XL4016_3
XL4016_4
XL4016_5
50V_470uF
25V_1000uF
T5A
T10A
+5V_1
+19V
Might change
this into a
more powerful
buck
24V_Main
This sheet focus only on power,
signal/control wires are not yet
include
Sheet: /OnBoard/
File: Onboard_Power.kicad_sch
Title:
Size: A4
Date:
Rev:
KiCad E.D.A. 9.0.6
Id: 5/4



Sheet: /OnBoard/		
File: Onboard_NAV_USER.kicad_sch		
Title:		
Size: A4	Date:	**Rev:**
KiCad E.D.A. 9.0.6		Id: 6/4